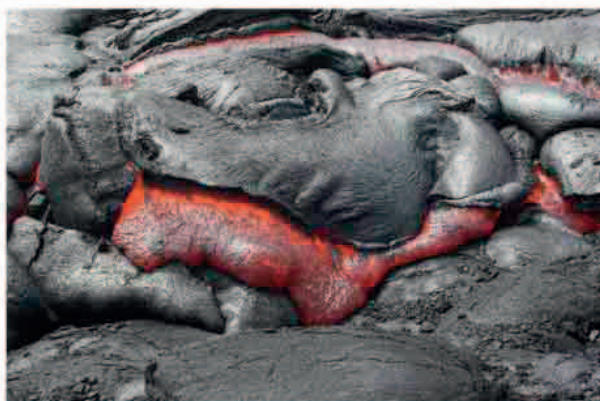


Igneous rocks

All rocks are mixtures of different minerals. This is obvious in igneous rocks such as granite, which are visibly made up of different types of crystals. These crystals form when molten magma or lava cools and freezes solid. As the crystals form, they lock together in a rigid mass with no gaps. This makes most igneous rocks extremely hard.

MAGMA AND LAVA

Although it's very hot, most of the mantle rock below Earth's crust is kept solid by intense pressure. However, if this pressure eases, or if the rock's melting point is lowered (by an added substance, such as water), the rock can melt. It becomes magma, which is less dense than solid mantle rock. The magma rises through the mantle and may slowly cool and solidify below ground, or erupt from volcanoes as liquid lava.

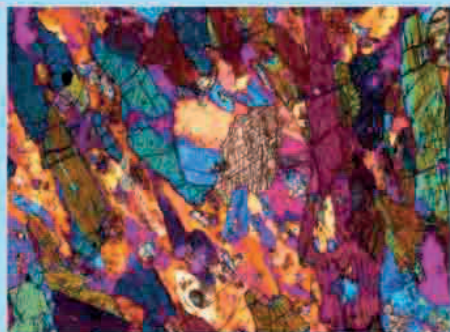


◀ MOLTEN ROCK

Red-hot lava spills down the side of a volcano in Hawaii. As it cools, it turns to basalt, one of the most common igneous rocks.

INTERLOCKING CRYSTALS

When a molten rock solidifies, each mineral ingredient forms its own crystals. As these grow, they become tightly interlocked in the very rigid structure that makes most igneous rocks so hard. This is clearly visible in this microscope image.



▲ LIGHT SHOW

Polarized light shining through this thin sample of a crystalline rock reveals the different mineral crystals in vivid colors.

CHANGING NATURES

When rock melts, some minerals may be left behind or added as it rises through the crust. This means that the mineral makeup of magma is different from the solid rock that produced it. In general, heavier elements, such as iron, are left behind. If the molten rock solidifies and then melts again, more changes occur. The heavier elements generally produce minerals that are darker in color, so each new stage is paler than the last. For example, granite, which is lightest in color, is created after basalt and diorite.



Basalt



Diorite



Granite

